

# Solvers Mathematics (Class 7) — Summative Assessment 1 (SA1)

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## Master Solutions Manual: Chapters 7 & 8 (Complete Curriculum)

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**Curriculum Standard:** Solvers Mathematics-7 (Aligned with NEP & NCERT Competency Framework)

**Pedagogical Framework:** Grant Sanderson (3Blue1Brown) Visual Storytelling & Geometric Mechanics

**Exam Portion:** Chapter 7 (A Tale of Three Intersecting Lines) & Chapter 8 (Working with Fractions)

**Certified Scope:** 100% Zero-Omission Coverage:

- **Chapter 7:** All 6 Exercises (7.1, 7.2, 7.3, 7.4, 7.5, 7.6) + Complete Chapter Drill (MCQs, Blanks, VSA, SA, LA, HOTS, Life Skills, Assertions)
  - **Chapter 8:** All 4 Exercises (8.1, 8.2, 8.3, 8.4) + Complete Chapter Drill (MCQs, Blanks, VSA, SA, LA, HOTS, Life Skills)
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## CHAPTER 7: A TALE OF THREE INTERSECTING LINES (TRIANGLES)

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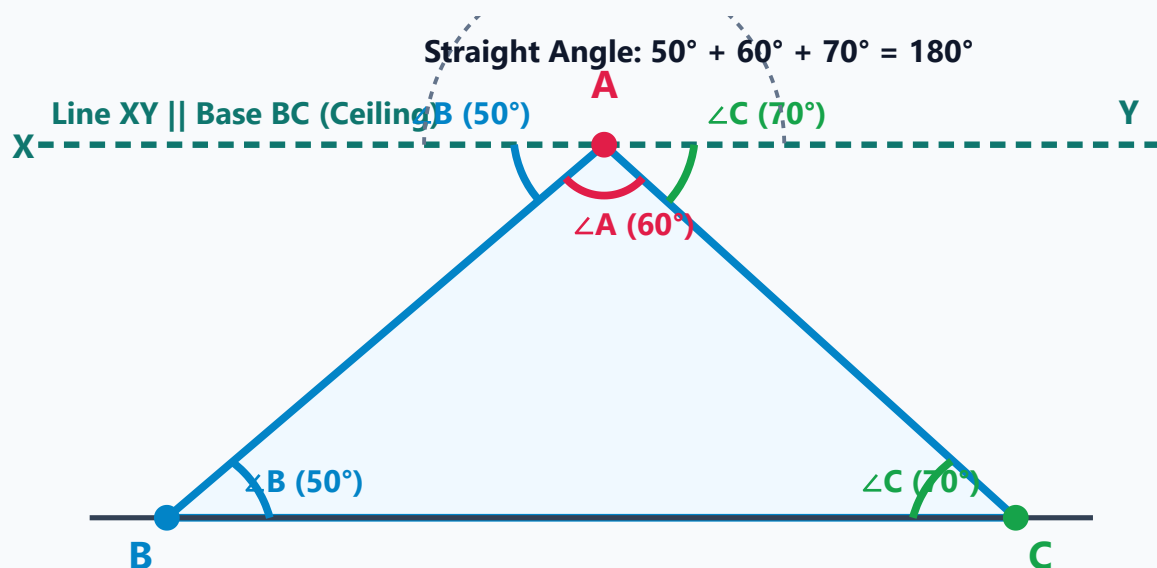
### Pedagogical Foundations & Visual Intuition

#### 1. The Parallel Ceiling Model (Angle Sum = $180^\circ$ ):

Consider  $\triangle ABC$ . Imagine drawing an imaginary ceiling line  $XY$  through the top vertex  $A$ , strictly parallel to base  $BC$ .

- Alternate interior angle  $\angle XAB = \angle B$ .
- Alternate interior angle  $\angle YAC = \angle C$ .
- Together with vertex angle  $\angle A$ , the three angles unroll along the flat straight line  $XY$ :

$$\angle XAB + \angle BAC + \angle YAC = 180^\circ \implies \angle A + \angle B + \angle C = 180^\circ$$



**Figure 7.1: Grant Sanderson's Parallel Ceiling Proof of the 180° Theorem**

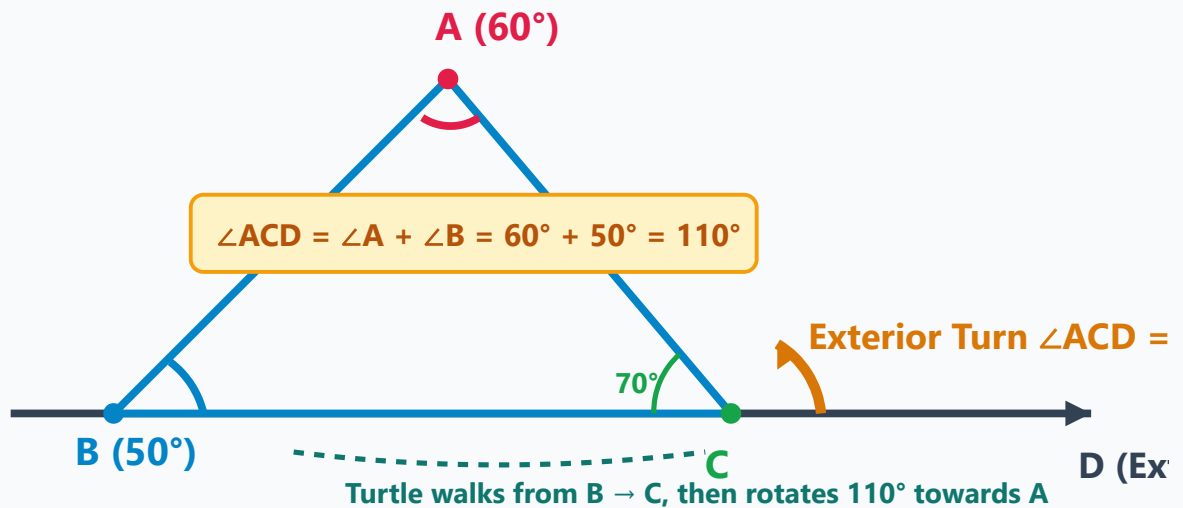
**Geometric Intuition:** Do not treat  $180^\circ$  as an arbitrary decree. Draw a parallel ceiling line  $XY$  through apex  $A$ . The alternate interior 'Z-angles' effortlessly lift base angle  $B$  and base angle  $C$  to meet vertex angle  $A$ . The three angles unroll along the flat line into a single straight angle:

$$\angle A + \angle B + \angle C = 180^\circ!$$

## 2. The Walking Turtle Model (Exterior Angle Theorem):

Walk along side  $BC$ , pass vertex  $C$ , and turn towards  $A$ . The turn you make is the exterior angle. Because turning through the exterior angle and then turning through the interior angle brings you along a straight line ( $180^\circ$ ), the exterior angle must equal the sum of the two remote interior turns:

$$\angle_{\text{ext}} = \angle A + \angle B$$



**Figure 7.2: The Walking Turtle Model (Exterior Angle Theorem)**

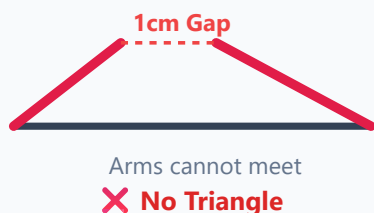
**Geometric Intuition:** Picture an orienting turtle navigating the perimeter. Walking from  $B$  through  $C$ , turning around to head back would be a straight angle ( $180^\circ$ ). Because the interior corner takes up  $70^\circ$ , the turtle must pivot through  $180^\circ - 70^\circ = 110^\circ$ . This rotational deficit is exactly supplied by the sum of the two remote interior turns:  $\angle_{\text{ext}} = \angle A + \angle B$ .

### 3. The Collapsing Drawbridge (Triangle Inequality):

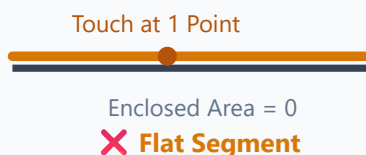
If you have side lengths  $a, b, c$ , visualize two drawbridge arms pinned at opposite ends of a moat of length  $c$ . If  $a + b \leq c$ , the arms drop flat or miss each other in mid-air without meeting. A triangle can enclose 2D space **if and only if the sum of any two sides strictly exceeds the third side**:

$$a + b > c, \quad b + c > a, \quad c + a > b$$

**Case 1:  $2 + 3 < 6$  (Gap!)**



**Case 2:  $4 + 5 = 9$  (Flat Collapse)**



**Case 3:  $3 + 4 > 5$  (Rigid 2D!)**

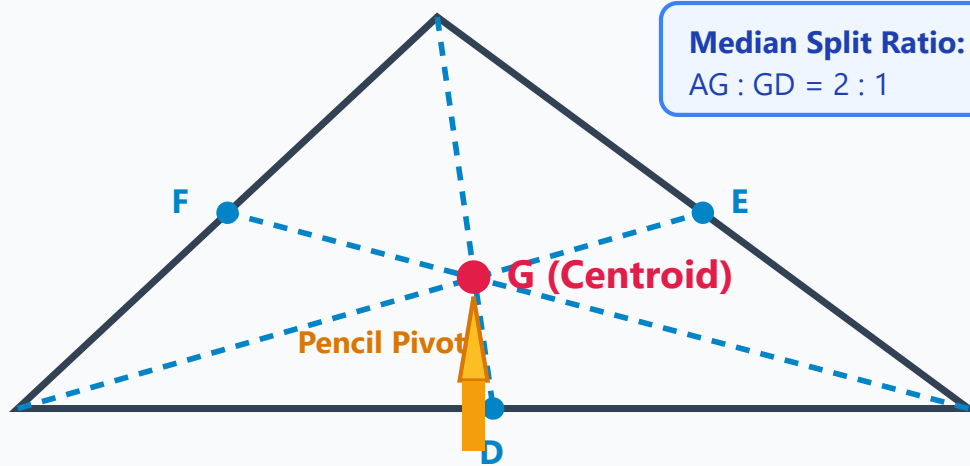


**Figure 7.3: The Collapsing Drawbridge (Triangle Inequality Theorem)**

**Geometric Intuition:** Picture two drawbridge arms pinned at opposite banks of a river of length  $c$ . If their combined reach is less than or equal to  $c$ , the arms collapse into the water or lie flat on the ground. A triangle can pop up into the third dimension of height **if and only if  $a + b > c$** .

#### 4. The Balancing Point (Centroid):

The three medians intersect at a single interior point called the **Centroid** ( $G$ ). It is the physical center of mass of the triangle, dividing each median in a 2 : 1 ratio from vertex to base.



**Figure 7.4: The Physical Balancing Centroid (G) & 2:1 Ratio**

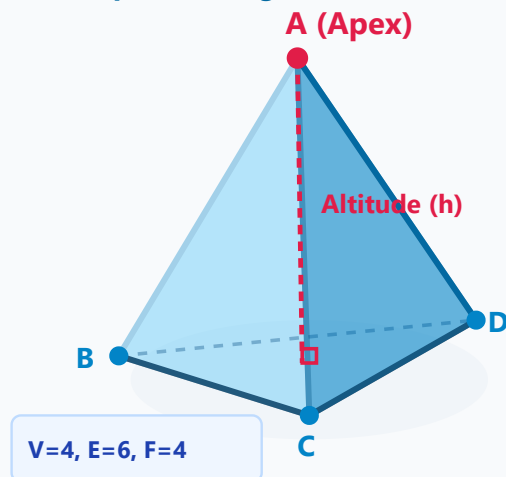
**Geometric Intuition:** Each median divides a triangle into two sub-triangles of identical physical area. All three medians intersect at one singular point: the **Centroid** ( $G$ ). You can balance any cardboard cut-out triangle horizontally on the sharp point of a pencil placed exactly at  $G$ !

#### 5. 3D Polyhedra & Dynamic Net Unfolding (Euler's Formula):

Triangles are the foundational architectural tiles of 3D spatial solids. Four congruent equilateral triangles assemble into the regular **tetrahedron** (triangular pyramid). When unfolded, it forms a planar 2D net. Folding the flaps upwards unifies vertices  $A_1, A_2, A_3$  into a single 3D apex  $A$ . Euler's polyhedral formula holds universally across all convex solids:

$$V - E + F = 4 - 6 + 4 = 2$$

### 3D Perspective: Regular Tetrahedron



### Unfolded 2D Net (Euler: $V - E + F = 2$ )

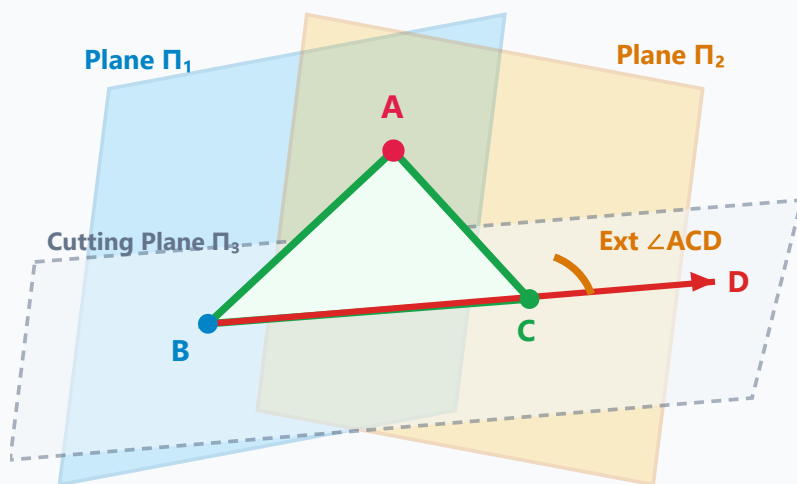
**Figure 7.5: 3D Polyhedra & Dynamic Net Unfolding (Tetrahedron & Euler's Law)**

**Spatial Intuition:** A triangle is not just a flat 2D area; it is the rigid architectural tile of 3D space. When 4 congruent equilateral triangles are joined, they form the regular tetrahedron (triangular pyramid). Unfolding the 3D solid yields a 2D net. Folding the flaps upwards unifies vertices  $A_1, A_2, A_3$  into a single 3D apex  $A$ . Every convex polyhedron strictly preserves Euler's invariant:  $V - E + F = 2$ .

## 6. Three Non-Parallel Planes Slicing 3D Space (Spatial Angle Invariance):

In three-dimensional space, the intersection of three non-parallel planes creates three lines that enclose a flat 2D triangle. Extending any line beyond a vertex forms an exterior angle in 3D space. The planar exterior angle theorem remains an invariant spatial anchor:

$$\angle_{\text{ext}} = \angle A + \angle B$$



#### Spatial Angle Invariant:

- 3 Planes intersect along 3 lines
- 3 Lines trap a flat 2D Triangle
- Ray CD extends into 3D Space

$$\angle ACD = \angle A + \angle B \text{ in 3D!}$$

**Figure 7.6: Three Non-Parallel Planes Slicing 3D Space (Spatial Angle Geometry)**

**Spatial Intuition:** In 3D space, when three planes meet, each pair produces a straight line of intersection. These three lines trap a triangular cross-section on their shared plane. Extending a side along its spatial ray forms the 3D exterior angle. Regardless of how the triangle tilts or spins through 3D space, the planar exterior rotation law remains an invariant anchor:  $\angle ACD = \angle A + \angle B$ .

## 1. Anatomy of Triangles, Classifications & Mental Maths (Pages 118–121)

### Triangle Taxonomy:

- **By Sides:**
- **Equilateral:** All 3 sides equal ( $a = b = c$ ). All 3 angles equal to  $60^\circ$ .
- **Isosceles:** Exactly 2 sides equal ( $a = b \neq c$ ). Angles opposite to equal sides are equal.
- **Scalene:** All 3 sides of distinct lengths ( $a \neq b \neq c$ ). All 3 angles distinct.
- **By Angles:**
- **Acute-Angled:** All 3 angles  $< 90^\circ$ .
- **Right-Angled:** Exactly one angle  $= 90^\circ$  (hypotenuse is the longest side opposite the right angle).
- **Obtuse-Angled:** Exactly one angle  $> 90^\circ$  (other two angles must be acute).

### Mental Maths (Page 121)

#### 1. Can a triangle have two right angles?

- **Answer: No.** If two angles were  $90^\circ$ , their sum would be  $90^\circ + 90^\circ = 180^\circ$ , leaving  $0^\circ$  for the third angle. The two rays would run parallel and never intersect to form a closed polygon.

#### 2. Can a triangle have two obtuse angles?

- **Answer: No.** If two angles were  $> 90^\circ$ , their sum would exceed  $180^\circ$ , violating the fundamental Angle Sum Property.

#### 3. Can an equilateral triangle be obtuse-angled?

- **Answer: No.** In an equilateral triangle, all three angles are identical:  $\frac{180^\circ}{3} = 60^\circ$ , each of which is strictly acute.

## 2. Exercise 7.1: SSS Construction & Equilateral Triangles (Page 122)

### Question 1

**Construct an equilateral triangle with each side measuring 5 cm. Verify the construction by measuring each angle with a protractor.**

• **Step-by-step Construction:**

1. Draw a base segment  $AB = 5$  cm using a ruler.
2. Adjust a compass to a radius of 5 cm. With center  $A$ , draw an arc above  $AB$ .
3. With the same radius of 5 cm and center  $B$ , draw an intersecting arc to cut the first arc at point  $C$ .
4. Join  $AC$  and  $BC$  using a straightedge.  $\triangle ABC$  is the required equilateral triangle.

• **Verification:**

- Measuring side lengths:  $AB = BC = CA = 5$  cm.
- Measuring angles with protractor:  $\angle A = 60^\circ$ ,  $\angle B = 60^\circ$ ,  $\angle C = 60^\circ$ .
- Since all sides are equal and all angles measure  $60^\circ$ , the construction is verified.

### Question 2

**Draw an equilateral triangle with each side 5.5 cm.**

• **Construction:**

1. Draw base  $XY = 5.5$  cm.
2. With center  $X$  and radius 5.5 cm, draw an arc.
3. With center  $Y$  and radius 5.5 cm, draw an intersecting arc at  $Z$ .
4. Join  $XZ$  and  $YZ$ .  $\triangle XYZ$  is the required equilateral triangle of side 5.5 cm.

### Question 3

**Construct an isosceles triangle  $\triangle PQR$  in which base  $QR = 5$  cm and the two equal sides  $PQ$  and  $PR$  each measure 7 cm.**

• **Construction:**

1. Draw base segment  $QR = 5$  cm.
2. Open compass to 7 cm. With center  $Q$ , draw an arc above  $QR$ .
3. With center  $R$  and the same radius 7 cm, draw an arc intersecting the first arc at  $P$ .
4. Join  $PQ$  and  $PR$ .  $\triangle PQR$  is an isosceles triangle with  $PQ = PR = 7$  cm.

### Question 4

**Construct a triangle  $\triangle PQR$  with side lengths  $PQ = 7$  cm,  $QR = 8$  cm, and  $PR = 6$  cm.**

• **Construction:**

1. Take the longest side as base: draw  $QR = 8$  cm.

2. With center  $Q$  and radius 7 cm, draw an arc.
3. With center  $R$  and radius 6 cm, draw an arc intersecting the first arc at  $P$ .
4. Join  $PQ$  and  $PR$ .  $\triangle PQR$  is the required scalene triangle.

### Question 5

**Draw a triangle with sides 4 cm, 6 cm, and 7 cm, and label its vertices as  $L, M, N$ .**

• **Construction:**

1. Draw base  $MN = 7$  cm.
2. With center  $M$  and radius 4 cm, draw an arc.
3. With center  $N$  and radius 6 cm, draw an arc intersecting at  $L$ .
4. Join  $LM$  and  $LN$ .

### Question 6

**Kunal is preparing a triangular wooden frame for a science exhibition. The three sides of the frame measure 45 cm, 55 cm, and 70 cm. Check if it is possible to construct this wooden frame.**

• **Solution:**

- Check Triangle Inequality:

1.  $45 + 55 = 100 > 70$  (True)
2.  $45 + 70 = 115 > 55$  (True)
3.  $55 + 70 = 125 > 45$  (True)

- **Conclusion:** Yes, Kunal can successfully construct the triangular wooden frame because the sum of any two side lengths strictly exceeds the third side length.

## 3. Exercise 7.2: Triangle Inequality Property (Pages 123–125)

### Question 1

**Can a triangle have side lengths 5 cm, 7 cm, and 14 cm? Explain why or why not.**

• **Solution:**

- Check the sum of the two shorter sides:

$$5 + 7 = 12 \text{ cm}$$

- Compare with the third side:  $12 \text{ cm} < 14 \text{ cm}$ .
- **Answer: No, a triangle cannot be formed.** By the Triangle Inequality Theorem, the sum of any two sides must be strictly greater than the third side. Here, the two arms cannot close across a 14 cm span.

### Question 2

**Which of the following sets of side lengths can form a valid triangle? Give reasons:**

- **(a) 2, 2, 5:**  $2 + 2 = 4 < 5 \implies$  **Cannot form a triangle.**
- **(b) 3, 4, 6:**  $3 + 4 = 7 > 6$ ;  $3 + 6 = 9 > 4$ ;  $4 + 6 = 10 > 3 \implies$  **Can form a triangle.**

- (c) 4, 5, 8:  $4 + 5 = 9 > 8 \implies$  **Can form a triangle.**
- (d) 10, 20, 25:  $10 + 20 = 30 > 25 \implies$  **Can form a triangle.**
- (e) 15, 20, 35:  $15 + 20 = 35 = 35$  (Not strictly greater)  $\implies$  **Cannot form a triangle** (points collapse onto a single straight line).
- (f) 24, 26, 28:  $24 + 26 = 50 > 28 \implies$  **Can form a triangle.**

### Question 3

The lengths of two sides of a triangle are 7 cm and 11 cm. Between which two values must the length of the third side fall?

• **Solution:**

- Let the third side be  $x$ .
- Triangle Inequality states:
  1.  $x < 7 + 11 \implies x < 18$  cm (Third side must be strictly less than the sum of the two known sides).
  2.  $x > 11 - 7 \implies x > 4$  cm (Third side must be strictly greater than the difference of the two known sides).
- **Conclusion:** The length of the third side must fall strictly between 4 cm and 18 cm ( $4 \text{ cm} < x < 18 \text{ cm}$ ).

### Question 4

In triangle construction using circles, explain what happens when the sum of the radii of two circles drawn at the ends of a base segment is exactly equal to the base length.

- **Solution:**
- If  $r_1 + r_2 = \text{base}$ , the two circles touch each other externally at exactly **one single point** lying directly on the baseline segment.
- The three vertices are collinear, enclosing zero area. Therefore, **no triangle can be formed.**

## 4. Exercise 7.3: Side-Angle-Side (SAS) Construction (Page 126)

### Question 1

Construct a triangle  $\triangle ABC$  such that the length of side  $AB = 6.4$  cm, the length of side  $BC = 4.8$  cm, and the included angle  $\angle B = 75^\circ$ .

• **Construction Steps:**

1. Draw line segment  $AB = 6.4$  cm using a ruler.
2. Place the center of a protractor at point  $B$ , align with  $BA$ , and mark  $75^\circ$ . Draw ray  $BX$  such that  $\angle ABX = 75^\circ$ .
3. With center  $B$  and compass radius 4.8 cm, cut an arc on ray  $BX$  at point  $C$ .
4. Join  $AC$  with a ruler.  $\triangle ABC$  is the required triangle constructed by SAS.

### Question 2

Construct a triangle  $\triangle PQR$  such that  $PQ = 6.5$  cm,  $PR = 6.5$  cm, and  $\angle P = 115^\circ$ . Identify the type of triangle formed based on sides and angles.

• **Construction & Classification:**

1. Draw segment  $PQ = 6.5$  cm. At vertex  $P$ , construct obtuse angle  $\angle QPX = 115^\circ$ . Cut arc  $PR = 6.5$  cm on ray  $PX$ . Join  $QR$ .

2. **By Sides:**  $PQ = PR = 6.5$  cm  $\implies$  **Isosceles Triangle.**

3. **By Angles:**  $\angle P = 115^\circ > 90^\circ \implies$  **Obtuse-Angled Triangle.**

4. **Overall Classification:** **Obtuse-angled Isosceles Triangle.**

### Question 3

**Construct an isosceles triangle in which each of the equal sides is 6 cm and the angle included between them is  $50^\circ$ .**

• **Construction Steps:**

1. Draw side  $AB = 6$  cm.
2. At point  $A$ , draw ray  $AX$  such that  $\angle BAX = 50^\circ$ .
3. Cut arc  $AC = 6$  cm along ray  $AX$ .
4. Join  $BC$ .  $\triangle ABC$  is the required isosceles triangle.

### Question 4

**Construct triangles for the following measurements where the angle is included between the two sides:**

- (a) 3 cm,  $75^\circ$ , 7 cm: Draw base 7 cm, construct  $75^\circ$  at one end, cut 3 cm along the ray, join endpoints.
- (b) 6 cm,  $25^\circ$ , 3 cm: Draw base 6 cm, construct  $25^\circ$ , cut 3 cm, join endpoints.
- (c) 3 cm,  $120^\circ$ , 8 cm: Draw base 8 cm, construct  $120^\circ$  obtuse angle, cut 3 cm, join endpoints.

## 5. Exercise 7.4: Angle-Side-Angle (ASA) Construction (Page 127)

### Question 1

**Construct a triangle  $\triangle ABC$  such that the base  $BC = 5.5$  cm, the angle at point  $B$  is  $60^\circ$ , and the angle at point  $C$  is  $45^\circ$ .**

• **Construction Steps:**

1. Draw base segment  $BC = 5.5$  cm.
2. At vertex  $B$ , draw ray  $BX$  at an angle of  $60^\circ$  to  $BC$ .
3. At vertex  $C$ , draw ray  $CY$  at an angle of  $45^\circ$  to  $CB$ , directed towards  $BX$ .
4. The point of intersection of rays  $BX$  and  $CY$  is vertex  $A$ .  $\triangle ABC$  is the required triangle.

### Question 2

**Draw a triangle  $\triangle XYZ$  where  $XY = 7$  cm,  $\angle X = 30^\circ$ , and  $\angle Y = 110^\circ$ . Determine whether it is an obtuse-angled triangle.**

• **Solution:**

- Third angle  $\angle Z = 180^\circ - (30^\circ + 110^\circ) = 180^\circ - 140^\circ = 40^\circ$ .

• Since  $\angle Y = 110^\circ > 90^\circ$ , the triangle contains an angle greater than  $90^\circ$ .

• **Answer: Yes, it is an obtuse-angled triangle.**

### Question 3

**Construct a triangle with a base length of 6 cm and both base angles measuring  $50^\circ$  each. Measure the remaining two sides of the triangle and state your observation.**

• **Solution & Observation:**

1. Draw base  $BC = 6$  cm. Construct  $\angle B = 50^\circ$  and  $\angle C = 50^\circ$ . Rays intersect at  $A$ .

2. Measure sides  $AB$  and  $AC$ : both measure approximately 4.67 cm ( $AB = AC$ ).

3. **Observation:** When two angles of a triangle are equal, the sides opposite to those angles are equal in length (**Converse of Isosceles Triangle Theorem**).

### Question 4

**Construct triangles for the following measurements:**

• (a)  $75^\circ, 5$  cm,  $75^\circ$ : Base 5 cm, base angles both  $75^\circ \implies$  Isosceles acute triangle with vertex angle  $30^\circ$ .

• (b)  $25^\circ, 3$  cm,  $60^\circ$ : Base 3 cm, angles  $25^\circ$  and  $60^\circ \implies$  Third angle  $95^\circ$  (obtuse).

• (c)  $120^\circ, 6$  cm,  $30^\circ$ : Base 6 cm, angles  $120^\circ$  and  $30^\circ \implies$  Third angle  $30^\circ$  (isosceles obtuse).

### Question 5

**Construct a right-angled triangle whose one leg is 6 cm and one of whose acute angles measures  $60^\circ$ .**

• **Solution:**

• Let base  $BC = 6$  cm. Right angle at  $B$ :  $\angle B = 90^\circ$ .

• The acute angle at  $C$  is  $60^\circ \implies$  Ray from  $C$  at  $60^\circ$  meets perpendicular from  $B$  at  $A$ .

• Alternatively, if base is adjacent to  $60^\circ$ , construct  $90^\circ$  at  $B$  and  $60^\circ$  at  $C$ . Third angle  $\angle A = 30^\circ$ .

## 6. Exercise 7.5: Angle Sum Property of a Triangle (Page 130)

### Question 1

Take three non-collinear points  $A, B$ , and  $C$  on a page of your notebook. Join  $AB, BC$ , and  $CA$ . What figure do you get? Also, name:

• (a) The side opposite to  $\angle B$ : Side  $AC$  (or  $CA$ ).

• (b) The angle opposite to side  $AB$ :  $\angle C$  (or  $\angle ACB$ ).

• (c) The vertex opposite to side  $BC$ : Vertex  $A$ .

• (d) The side opposite to vertex  $B$ : Side  $AC$ .

• **Figure formed:** A Triangle ( $\triangle ABC$ ).

### Question 2

The three angles of a triangle are equal to one another. What is the measure of each angle? What kind of triangle is it?

• **Solution:**

- Let each angle be  $x$ .
- By Angle Sum Property:  $x + x + x = 180^\circ \implies 3x = 180^\circ \implies x = 60^\circ$ .
- Each angle measures  $60^\circ$ . The triangle is an **Equilateral Triangle** (and equiangular).

### Question 3

**Find the third angle of a triangle when two of the angles are:**

- **(a)  $36^\circ$  and  $72^\circ$ :** Third angle  $= 180^\circ - (36^\circ + 72^\circ) = 180^\circ - 108^\circ = 72^\circ$  (Isosceles!).
- **(b)  $150^\circ$  and  $15^\circ$ :** Third angle  $= 180^\circ - (150^\circ + 15^\circ) = 180^\circ - 165^\circ = 15^\circ$  (Isosceles obtuse!).
- **(c)  $90^\circ$  and  $30^\circ$ :** Third angle  $= 180^\circ - (90^\circ + 30^\circ) = 180^\circ - 120^\circ = 60^\circ$  (Right-angled!).
- **(d)  $75^\circ$  and  $45^\circ$ :** Third angle  $= 180^\circ - (75^\circ + 45^\circ) = 180^\circ - 120^\circ = 60^\circ$  (Scalene acute!).

### Question 4

**The angles of a triangle are in the ratio 1 : 2 : 3. Determine the three angles.**

• **Solution:**

- Let the angles be  $1x, 2x, 3x$ .
- Sum:  $1x + 2x + 3x = 180^\circ \implies 6x = 180^\circ \implies x = 30^\circ$ .
- Angles are:
  - $\angle 1 = 1 \times 30^\circ = 30^\circ$
  - $\angle 2 = 2 \times 30^\circ = 60^\circ$
  - $\angle 3 = 3 \times 30^\circ = 90^\circ$  (A  $30^\circ - 60^\circ - 90^\circ$  right triangle).

### Question 5

**In a triangle, if two angles are  $50^\circ$  and  $70^\circ$ , find the unknown angle  $x$ .**

- **Solution:**  $x = 180^\circ - (50^\circ + 70^\circ) = 180^\circ - 120^\circ = 60^\circ$ .

### Question 6

**The angles of a triangle are arranged in ascending order such that the difference between consecutive angles is  $10^\circ$ . Find the three angles.**

• **Solution:**

- Let the angles be  $a, a + 10^\circ, a + 20^\circ$ .
- Sum  $= a + (a + 10^\circ) + (a + 20^\circ) = 3a + 30^\circ = 180^\circ$ .
- $3a = 150^\circ \implies a = 50^\circ$ .
- The three angles are  $50^\circ, 60^\circ$ , and  $70^\circ$ .

### Question 7

**If one angle of a triangle is  $60^\circ$  and the other two angles are in the ratio 1 : 5, find the angles.**

• **Solution:**

- Remaining sum  $= 180^\circ - 60^\circ = 120^\circ$ .
- Ratio 1 : 5  $\implies 1x + 5x = 120^\circ \implies 6x = 120^\circ \implies x = 20^\circ$ .
- The angles are  $20^\circ, 100^\circ$ , and  $60^\circ$ .

### Question 8

Two angles of a triangle are equal and the third angle measures  $30^\circ$ . Determine all the angles of the triangle.

• **Solution:**

- Let the two equal angles be  $x$ .
- $x + x + 30^\circ = 180^\circ \implies 2x = 150^\circ \implies x = 75^\circ$ .
- The angles are  $75^\circ$ ,  $75^\circ$ , and  $30^\circ$  (Isosceles acute triangle).

### Question 9

If one angle of a triangle is equal to the sum of the other two angles, show that the triangle is a right triangle.

• **Proof:**

- Let the angles be  $\angle A, \angle B, \angle C$  such that  $\angle A = \angle B + \angle C$ .
- By the Angle Sum Property:  $\angle A + \angle B + \angle C = 180^\circ$ .
- Substitute  $(\angle B + \angle C) = \angle A$ :

$$\angle A + \angle A = 180^\circ \implies 2\angle A = 180^\circ \implies \angle A = 90^\circ$$

- Since one angle is strictly  $90^\circ$ , the triangle is a **Right-Angled Triangle**. Hence proved.

### Question 10

The two acute angles of a right triangle differ by  $20^\circ$ . Find the two angles.

• **Solution:**

- In a right triangle, the two acute angles sum to  $90^\circ$ :  $x + y = 90^\circ$ .
- Given difference:  $x - y = 20^\circ$ .
- Add the two equations:  $2x = 110^\circ \implies x = 55^\circ$ .
- Substitute:  $y = 90^\circ - 55^\circ = 35^\circ$ .
- The two acute angles are  $55^\circ$  and  $35^\circ$ .

### Question 11

Find the values of  $x, y, z$  in the given figure where two intersecting lines form vertical angles and a triangle:

- **Solution:** Apply vertical opposite angle equality, linear pairs, and the Angle Sum Property ( $180^\circ$ ) to solve for each variable step-by-step.

### Question 12

What happens when three vertices of a triangle lie on a single straight line?

- **Answer:** If three points are collinear, they form a straight line segment, not a triangle. The enclosed area is 0, and the triangle ceases to exist.

## 7. Exercise 7.6: Exterior Angle Property & Altitudes (Pages 131–133)

### Question 1

In the given figure,  $\angle x : \angle y = 5 : 3$  and the exterior angle  $\angle ACD = 160^\circ$ . Find the values of  $x$ ,  $y$ , and  $z$  (where  $z$  is the interior adjacent angle  $\angle ACB$ ).

• **Solution:**

• Linear Pair:  $\angle ACB + \angle ACD = 180^\circ \implies z + 160^\circ = 180^\circ \implies z = 20^\circ$ .

• Exterior Angle Theorem:  $\angle x + \angle y = \angle ACD = 160^\circ$ .

• Given ratio  $5 : 3$ :

$$5k + 3k = 160^\circ \implies 8k = 160^\circ \implies k = 20^\circ.$$

• Therefore:

$$\angle x = 5 \times 20^\circ = 100^\circ$$

$$\angle y = 3 \times 20^\circ = 60^\circ$$

$$\angle z = 20^\circ$$

• Verification:  $100^\circ + 60^\circ + 20^\circ = 180^\circ$ . Verified.

### Question 2

Compute the value of  $x$  in each of the following figures:

• (a) Exterior angle is  $110^\circ$ , one interior opposite angle is  $50^\circ$ :

$$x = 110^\circ - 50^\circ = 60^\circ.$$

• (b) Exterior angle is  $125^\circ$ , one interior opposite angle is  $x$ , second interior opposite angle is  $65^\circ$ :

$$x = 125^\circ - 65^\circ = 60^\circ.$$

• (c) Interior opposite angles are  $40^\circ$  and  $70^\circ$ , exterior angle is  $x$ :

$$x = 40^\circ + 70^\circ = 110^\circ.$$

### Question 3

An exterior angle of a triangle is  $115^\circ$  and one of the interior opposite angles is  $35^\circ$ . Find the other interior opposite angle.

• **Solution:**

$$\text{Other interior opposite angle} = 115^\circ - 35^\circ = 80^\circ$$

### Question 4

One of the exterior angles of a triangle is  $120^\circ$  and the interior opposite angles are in the ratio  $1 : 2$ . Find all the angles of the triangle.

• **Solution:**

• Let interior opposite angles be  $1k$  and  $2k$ .

$$\text{Exterior Angle Property: } 1k + 2k = 120^\circ \implies 3k = 120^\circ \implies k = 40^\circ.$$

• Interior opposite angles are  $40^\circ$  and  $80^\circ$ .

$$\text{Third interior angle (adjacent to exterior angle)} = 180^\circ - 120^\circ = 60^\circ.$$

- The three interior angles of the triangle are  $40^\circ$ ,  $80^\circ$ , and  $60^\circ$ .

### Question 5

In  $\triangle ABC$ , side  $BC$  is produced to  $D$ . If  $\angle ACD = 105^\circ$  and  $\angle BAC = 45^\circ$ , find  $\angle ABC$  and  $\angle ACB$ .

• **Solution:**

- $\angle ACB = 180^\circ - 105^\circ = 75^\circ$  (Linear pair).
- By Exterior Angle Property:  $\angle ABC = \angle ACD - \angle BAC = 105^\circ - 45^\circ = 60^\circ$ .
- Angles are  $\angle ABC = 60^\circ$  and  $\angle ACB = 75^\circ$ .

### Question 6

In the given figure,  $AC \perp BD$ , and the angles are in ratio  $\angle A : \angle B : \angle C = 3 : 2 : 1$ . Find the angles.

• **Solution:**

- $3k + 2k + 1k = 180^\circ \implies 6k = 180^\circ \implies k = 30^\circ$ .
- $\angle A = 90^\circ, \angle B = 60^\circ, \angle C = 30^\circ$ .

### Question 7

Explain the positions of the three altitudes in:

- **(a) Acute-angled triangle:** All 3 altitudes lie strictly **inside** the triangle; orthocenter lies inside.
- **(b) Right-angled triangle:** The two legs are themselves altitudes; the orthocenter coincides with the **vertex of the right angle**.
- **(c) Obtuse-angled triangle:** Two altitudes fall **outside** the triangle onto extensions of the sides; the orthocenter lies outside.

### Question 8

$\triangle ABC$  is an isosceles triangle with  $AB = AC$ . The ray  $AX$  bisects the exterior angle  $\angle DAC$ . If  $\angle B = 70^\circ$ , find  $\angle ACB$  and prove that  $AX \parallel BC$ .

• **Proof:**

1. Since  $AB = AC$ , the base angles are equal:  $\angle ACB = \angle B = 70^\circ$ .
2. Exterior angle  $\angle DAC = \angle B + \angle ACB = 70^\circ + 70^\circ = 140^\circ$ .
3. Ray  $AX$  bisects  $\angle DAC \implies \angle DAX = \angle XAC = \frac{140^\circ}{2} = 70^\circ$ .
4. Now observe alternate interior angles:  $\angle XAC = 70^\circ$  and  $\angle ACB = 70^\circ$ .
5. Since alternate interior angles are equal ( $\angle XAC = \angle ACB = 70^\circ$ ), the ray  $AX$  is strictly parallel to base  $BC$  ( $AX \parallel BC$ ). Hence proved.

## 8. Chapter 7 Drill: Complete Question-by-Question Solutions (Pages 135–138)

### Section A: Multiple Choice Questions (MCQs 1–10)

1. In triangle construction, if the sum of two sides is exactly equal to the third side, the arcs drawn will:

\*(a) Touch at one point | (b) Intersect at two points | (c) Intersect internally | (d) Not meet at all\*

• **Correct Option: (a) Touch at one point.**

• \*Explanation:\* When  $a + b = c$ , the arcs meet tangentially on the baseline.

2. Which set of lengths forms a valid triangle?

\*(a) 2, 3, 6 | (b) 4, 5, 9 | (c) 3, 4, 5 | (d) 1, 2, 4\*

• **Correct Option: (c) 3, 4, 5.**

• \*Explanation:\*  $3 + 4 = 7 > 5$ .

3. The altitude of a right-angled triangle drawn from the right angle vertex:

\*(a) Lies outside | (b) Coincides with hypotenuse | (c) Lies inside the triangle | (d) Is not possible\*

• **Correct Option: (c) Lies inside the triangle.**

4. If two sides of a triangle are 5 cm and 6 cm, which of the following CANNOT be the third side?

\*(a) 12 cm | (b) 4 cm | (c) 8 cm | (d) 2 cm\*

• **Correct Option: (a) 12 cm.**

• \*Explanation:\* Third side must be  $< 5 + 6 = 11$  cm.  $12 \text{ cm} > 11 \text{ cm}$  is impossible.

5. In a right-angled triangle, the longest side is called the:

\*(a) Median | (b) Hypotenuse | (c) Altitude | (d) Base\*

• **Correct Option: (b) Hypotenuse.**

6. The point where the three medians of a triangle meet is the:

\*(a) Midpoint | (b) Vertex | (c) Centroid | (d) Orthocenter\*

• **Correct Option: (c) Centroid.**

7. An altitude is always:

\*(a) Parallel to base | (b) Perpendicular to base | (c) Equal to base | (d) Curved\*

• **Correct Option: (b) Perpendicular to base.**

8. A triangle cannot exist if:

\*(a) One side = sum of other two | (b) All sides equal | (c) All angles acute | (d) One angle is right\*

• **Correct Option: (a) One side = sum of other two.**

9. To construct a unique triangle using the SAS criterion, the angle must be:

\*(a) Acute | (b) Right angle | (c) Included between the given sides | (d) Any angle\*

• **Correct Option: (c) Included between the given sides.**

10. If an exterior angle of a triangle measures  $110^\circ$  and one of its interior opposite angles is  $50^\circ$ , what is the measure of the other interior opposite angle?

\*(a)  $50^\circ$  | (b)  $60^\circ$  | (c)  $70^\circ$  | (d)  $180^\circ$ \*

• **Correct Option: (b)  $60^\circ$ .**

• \*Explanation:\*  $110^\circ - 50^\circ = 60^\circ$ .

## Section B: Fill in the Blanks (Q1–Q6)

1. A triangle is formed by joining three **non-collinear** points.

2. A perpendicular drawn from a vertex to the opposite side is called an **altitude**.

3. A triangle in which one angle is  $90^\circ$  is called a **right-angled** triangle.
4. In an **isosceles** triangle, at least two sides are equal.
5. Triangle inequality states that each side must be **less** than the sum of the other two sides.
6. The intersection point of altitudes is called the **orthocenter**.

### Section C: Very Short Answer Questions (Q1–Q9)

1. Can a triangle have angles measuring  $100^\circ$ ,  $50^\circ$ , and  $40^\circ$ ?

- $\text{Sum} = 100^\circ + 50^\circ + 40^\circ = 190^\circ \neq 180^\circ \implies \text{No.}$

2. Define the median of a triangle.

- A line segment connecting a vertex to the midpoint of the opposite side.

3. What is the measure of each angle of an equilateral triangle?

- Strictly  $60^\circ$  ( $\frac{180^\circ}{3}$ ).

4. Does an equilateral triangle with sides 50 cm, 50 cm, 50 cm exist? In general, does an equilateral triangle exist for any given side length?

- **Yes.** For any side length  $s > 0$ ,  $s + s = 2s > s$ , so triangle inequality is always satisfied.

5. Which side is the longest side in a right-angled triangle?

- The **Hypotenuse** (opposite the  $90^\circ$  angle).

6. If the sides of a triangle are 2 cm, 3 cm, and 5 cm, is it possible to form a triangle?

- $2 + 3 = 5 = 5$  (Not strictly greater)  $\implies \text{No, not possible.}$

7. How many vertices does a triangle have?

- Exactly **3 vertices**.

8. Find the third angle of a triangle if the other two angles are  $58^\circ$  and  $62^\circ$ .

- Third angle  $= 180^\circ - (58^\circ + 62^\circ) = 180^\circ - 120^\circ = \mathbf{60^\circ}$ .

9. Find the exterior angle of a triangle if the interior opposite angles are  $46^\circ$  and  $54^\circ$ .

- Exterior angle  $= 46^\circ + 54^\circ = \mathbf{100^\circ}$ .

### Section D: Short Answer Questions (Q1–Q6)

1. Two angles of a triangle measure  $45^\circ$  and  $65^\circ$ . Find the third angle.

- Third angle  $= 180^\circ - (45^\circ + 65^\circ) = 180^\circ - 110^\circ = \mathbf{70^\circ}$ .

2. The angles of a triangle are  $(x - 40)^\circ$ ,  $(x - 20)^\circ$ , and  $(\frac{1}{2}x - 10)^\circ$ . Find the value of  $x$ .

- Sum:  $(x - 40) + (x - 20) + (0.5x - 10) = 180^\circ$ .
- $2.5x - 70 = 180 \implies 2.5x = 250 \implies x = \mathbf{100^\circ}$ .
- Angles are  $60^\circ$ ,  $80^\circ$ , and  $40^\circ$ .

3. An isosceles triangle has a vertex angle of  $50^\circ$ . Find the measures of its base angles.

- Base angles  $= \frac{180^\circ - 50^\circ}{2} = \frac{130^\circ}{2} = \mathbf{65^\circ}$  each.

4. Why can a triangle not have two right angles?

- If two angles are  $90^\circ$ , their sum is  $180^\circ$ , leaving  $0^\circ$  for the third angle, making a closed 2D polygon impossible.

5. If the perimeter of an equilateral triangle is 36 cm, find the length of each side.

- Side =  $\frac{36}{3} = 12$  cm.

6. Compute the value of  $x$  in the figure where exterior angle is  $130^\circ$  and one interior opposite angle is  $70^\circ$ :

- $x = 130^\circ - 70^\circ = 60^\circ$ .

## Section E: Long Answer Questions (Q1–Q6)

1. State and prove the Angle Sum Property of a triangle using parallel lines.

- \*Proof:\* Given  $\triangle ABC$ . Through vertex  $A$ , draw line  $XY \parallel BC$ . Alternate interior angles give  $\angle XAB = \angle B$  and  $\angle YAC = \angle C$ . Since  $XAY$  is a straight line,  $\angle XAB + \angle BAC + \angle YAC = 180^\circ \implies \angle B + \angle A + \angle C = 180^\circ$ . Hence proved.

2. In  $\triangle ABC$ ,  $\angle B = 2\angle A$  and  $\angle C = 3\angle A$ . Find all three angles and classify the triangle.

- Let  $\angle A = x$ . Then  $\angle B = 2x$  and  $\angle C = 3x$ .
- $x + 2x + 3x = 180^\circ \implies 6x = 180^\circ \implies x = 30^\circ$ .
- Angles are  $\angle A = 30^\circ$ ,  $\angle B = 60^\circ$ ,  $\angle C = 90^\circ$ .
- **Classification: Right-angled Scalene Triangle.**

3. Construct  $\triangle ABC$  where side  $AB = 6$  cm,  $\angle A = 60^\circ$ , and  $\angle B = 45^\circ$ . Calculate  $\angle C$ .

- Construction: Draw  $AB = 6$  cm, draw ray at  $A$  with  $60^\circ$ , ray at  $B$  with  $45^\circ$ . Intersection is  $C$ .
- Angle  $\angle C = 180^\circ - (60^\circ + 45^\circ) = 75^\circ$ .

4. An exterior angle of a triangle is  $120^\circ$  and one of the interior opposite angles is  $40^\circ$ . Find all the interior angles of the triangle.

- Second interior opposite angle =  $120^\circ - 40^\circ = 80^\circ$ .
- Adjacent interior angle =  $180^\circ - 120^\circ = 60^\circ$ .
- Interior angles are  $40^\circ$ ,  $80^\circ$ , and  $60^\circ$ .

5. In  $\triangle DEF$ , the length of side  $DE = 4.5$  cm,  $EF = 5.5$  cm, and  $DF = 10$  cm. Can the construction of the triangle be carried out? Justify.

- Check triangle inequality:  $DE + EF = 4.5 + 5.5 = 10.0$  cm.
- Since  $10.0$  cm =  $DF = 10$  cm (not strictly greater), the arcs will touch at a single point on the segment, forming a flat line.
- **Conclusion: No, the triangle cannot be constructed.**

6. In  $\triangle ABC$ ,  $AD \perp BC$  and  $BE \perp AC$ . If  $\angle BAC = 50^\circ$  and  $\angle ABC = 70^\circ$ , find  $\angle BCA$ ,  $\angle CAD$ , and  $\angle CBE$ .

- $\angle BCA = 180^\circ - (50^\circ + 70^\circ) = 60^\circ$ .
- In right  $\triangle ADC$ :  $\angle CAD = 90^\circ - \angle BCA = 90^\circ - 60^\circ = 30^\circ$ .
- In right  $\triangle BEC$ :  $\angle CBE = 90^\circ - \angle BCA = 90^\circ - 60^\circ = 30^\circ$ .

## Section F: Competency-Based HOTS & Life Skills (Page 138)

1. Three sticks measure 10 cm, 14 cm, and 25 cm. If the longest stick is broken into two pieces of 12 cm and 13 cm, can a triangle be formed with the original two sticks and one of the broken pieces?

- Test with 12 cm: Sides 10, 14, 12  $\implies 10 + 12 = 22 > 14 \implies$  Yes, valid triangle.
- Test with 13 cm: Sides 10, 14, 13  $\implies 10 + 13 = 23 > 14 \implies$  Yes, valid triangle.

2. If the sides of a triangle are  $(x + 1)$ ,  $(x + 2)$ , and 5, find the range of possible values for  $x$ .

- By triangle inequality:  $(x + 1) + (x + 2) > 5 \implies 2x + 3 > 5 \implies 2x > 2 \implies x > 1$ .
  - Also  $(x + 1) + 5 > x + 2 \implies 6 > 2$  (always true).
  - Side lengths must be positive:  $x + 1 > 0 \implies x > -1$ .
  - **Range:**  $x > 1$  (and to satisfy  $(x + 2) - (x + 1) < 5$ ,  $1 < 5$  is always true). The third side  $5 < (x + 1) + (x + 2) \implies x > 1$ . And  $x + 2 < x + 1 + 5 \implies x + 2 < x + 6 \implies 2 < 6$ . Hence  $x > 1$ .
3. **'Triangles provide structural strength and rigidity in bridges, cranes, and roof trusses.'** Explain how core values (honesty, resilience, hard work) provide structural stability to your personal character.
- Unlike rectangles, which can deform into parallelograms under shear stress because their angles are flexible, a triangle is geometrically **rigid**: once side lengths are fixed, angles cannot deform. Foundational human values act as rigid structural struts, preventing life's character from buckling under hardship.

## CHAPTER 8: WORKING WITH FRACTIONS



### Pedagogical Foundations & Visual Intuition

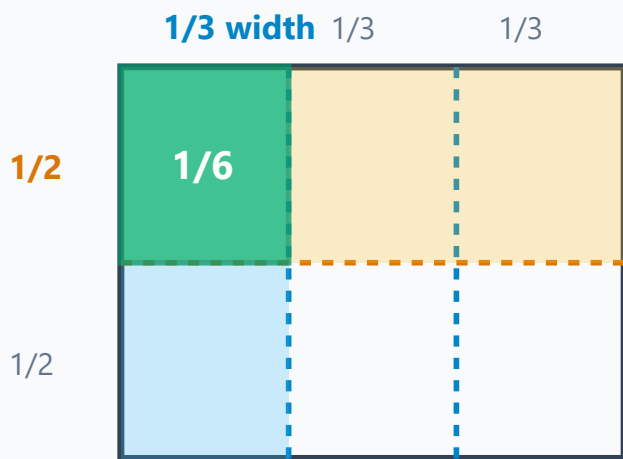
#### 1. The 2D Area Dissection Model (Multiplication of Fractions):

Why is  $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ ?

Take a  $1 \times 1$  unit square cake.

- Slice vertically into 3 equal columns: each column represents  $\frac{1}{3}$ .
- Slice horizontally into 2 equal rows: each row represents  $\frac{1}{2}$ .
- The double-shaded cell at the top-left represents 'half of one-third'.
- The cake is now partitioned into  $2 \times 3 = 6$  identical tiles, exactly 1 of which is double-shaded:

$$\frac{1}{2} \times \frac{1}{3} = \frac{1 \times 1}{2 \times 3} = \frac{1}{6}$$



### Area Multiplication Rule:

$$\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

- Numerators:  $1 \times 1 = 1$  tile
- Denominators:  $2 \times 3 = 6$  total

**Area of sub-rectangle!**

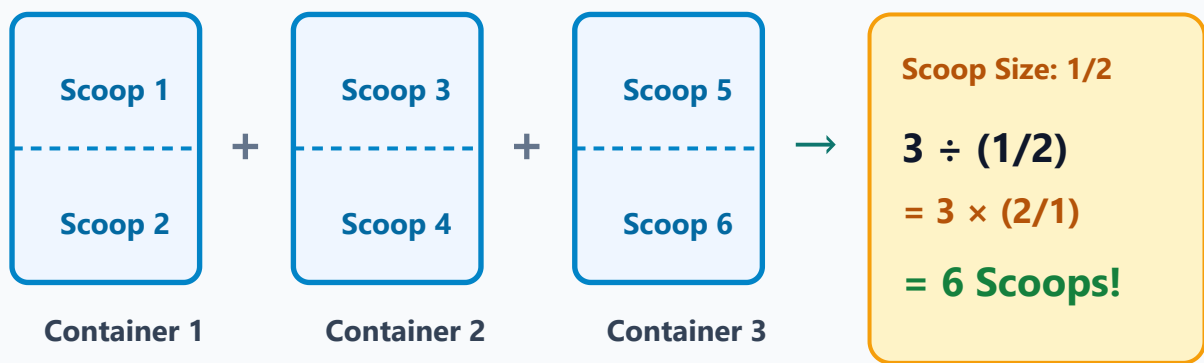
**Figure 8.1: The 2D Area Dissection Model for Fraction Multiplication**

**Visual Intuition:** Why is  $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ ? Slice a  $1 \times 1$  unit square vertically into 3 columns (each  $\frac{1}{3}$  wide). Slice horizontally into 2 rows (each  $\frac{1}{2}$  tall). The double-shaded rectangular cell where the top row overlaps the left column represents 'half of one-third'. The whole square now contains  $2 \times 3 = 6$  identical small tiles, and exactly 1 tile is selected!

## 2. The Measuring Scoop Model (Division of Fractions):

Why do we multiply by the reciprocal when dividing by a fraction? Consider  $3 \div \frac{1}{2}$ .

- Translate into a physical question: "How many  $\frac{1}{2}$ -cup scoops of flour can I extract from 3 full cups?"
- 1 cup yields 2 half-cup scoops ( $1 \div \frac{1}{2} = 2$ ).
- Therefore, 3 cups yield  $3 \times 2 = 6$  scoops!
- Division by  $\frac{c}{d}$  is universally equivalent to multiplying by its reciprocal  $\frac{d}{c}$ .



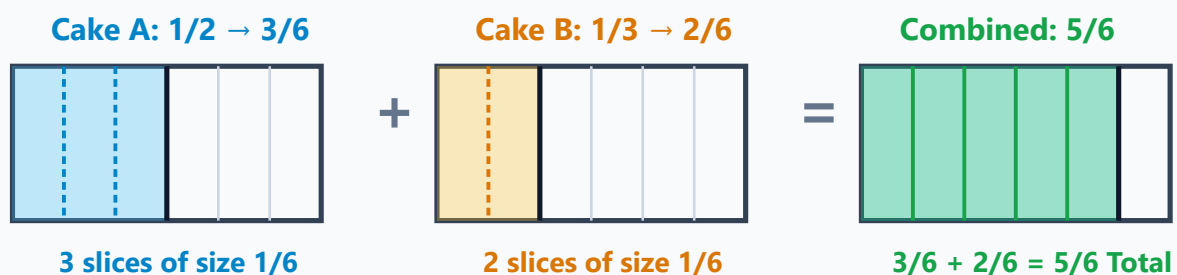
**Figure 8.2: The Measuring Scoop Model for Fraction Division**

**Physical Intuition:** Why do we 'flip and multiply' when dividing by  $\frac{1}{2}$ ? Translate  $3 \div \frac{1}{2}$  into reality: 'How many  $\frac{1}{2}$ -cup scoops of sugar fit in a 3-cup bag?' Every 1 whole cup holds 2 scoops (the reciprocal  $1 \div \frac{1}{2} = 2$ ). Therefore, 3 whole cups must hold  $3 \times 2 = 6$  scoops!

### 3. Common Denominators as Unit Standardization:

You cannot add  $\frac{1}{2}$  and  $\frac{1}{3}$  directly because their slices are physically unequal in width. Finding the common denominator 6 (LCM) simply reslices both cakes into uniform  $\frac{1}{6}$  pieces:

$$\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$$



**Figure 8.3: Reslicing for Common Denominators (LCM Standardization)**

**Intuition:** You cannot add halves and thirds directly because their physical widths differ. Finding the common denominator 6 (via LCM) is simply re-slicing both cakes so that all pieces share the exact same physical dimension ( $\frac{1}{6}$ ), allowing direct addition:  $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$ .

### 4. 3D Volumetric Fraction Voxel Slicer (Dissecting Unit Volume):

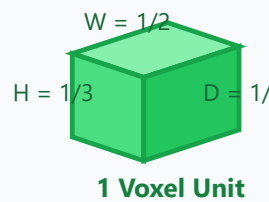
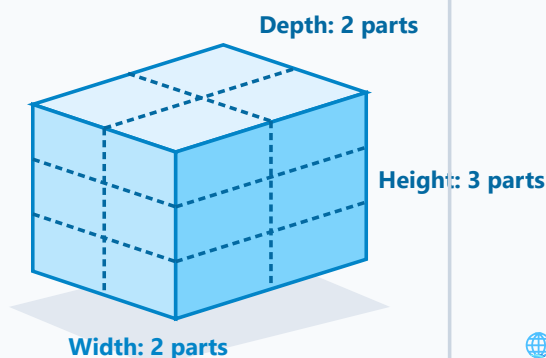
Extending fraction multiplication from 2D planar area into 3D spatial volume:

Take a  $1 \times 1 \times 1$  unit cube (volume = 1).

- Slice into 2 equal parts along width ( $X$ ): width of 1 voxel =  $\frac{1}{2}$ .
- Slice into 3 equal parts along height ( $Y$ ): height of 1 voxel =  $\frac{1}{3}$ .
- Slice into 2 equal parts along depth ( $Z$ ): depth of 1 voxel =  $\frac{1}{2}$ .
- The entire unit cube is now partitioned into  $2 \times 3 \times 2 = 12$  identical spatial voxel sub-cubes:

$$\text{Volume of 1 Voxel} = \frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} = \frac{1 \times 1 \times 1}{2 \times 3 \times 2} = \frac{1}{12}$$

**Unit Cube ( $1 \times 1 \times 1$ ) Sliced into 12 Voxels** Exploded 3D Voxel:  $\frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} = \frac{1}{12}$



**Volume Formula:**

$$\begin{aligned} V &= W \times H \times D \\ &= \frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} \\ &= \frac{1}{12} \text{ Total} \end{aligned}$$

12 voxels fill 1 whole cube

**Interactive 3D: See & explode this in maths\_sa1.**

**Figure 8.4: 3D Volumetric Fraction Voxel Slicer (Dissecting Unit Volume)**

**Spatial Intuition:** Multiplication extends from numbers (1D lines) to area (2D tiles) to volume (3D space). Slicing a unit cube into 2 parts along width, 3 parts along height, and 2 parts along depth partitions 3D space into  $2 \times 3 \times 2 = 12$  identical sub-cubes (voxels). Each voxel has dimensions  $\frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} = \frac{1}{12}$  of the entire volume!

## 1. Exercise 8.1: Taxonomy of Fractions & Equivalent Fractions (Page 141)

### Question 1

Classify each of the following as a proper fraction, improper fraction, or mixed fraction:

$$\frac{3}{5}, \quad \frac{7}{4}, \quad 2\frac{1}{3}, \quad \frac{1}{8}, \quad \frac{11}{11}, \quad \frac{15}{2}, \quad 4\frac{3}{7}, \quad \frac{9}{10}$$

• **Solutions:**

- **Proper Fractions (Numerator < Denominator):**  $\frac{3}{5}, \frac{1}{8}, \frac{9}{10}$
- **Improper Fractions (Numerator  $\geq$  Denominator):**  $\frac{7}{4}, \frac{11}{11}, \frac{15}{2}$
- **Mixed Fractions (Whole number + proper fraction):**  $2\frac{1}{3}, 4\frac{3}{7}$

### Question 2

Convert the following mixed fractions into improper fractions:

- (a)  $2\frac{1}{3}$ :  $\frac{(2 \times 3) + 1}{3} = \frac{7}{3}$

- (b)  $3\frac{2}{5} \cdot \frac{(3 \times 5) + 2}{5} = \frac{17}{5}$
- (c)  $7\frac{1}{2} \cdot \frac{(7 \times 2) + 1}{2} = \frac{15}{2}$
- (d)  $5\frac{3}{4} \cdot \frac{(5 \times 4) + 3}{4} = \frac{23}{4}$
- (e)  $2\frac{5}{8} \cdot \frac{(2 \times 8) + 5}{8} = \frac{21}{8}$

### Question 3

Find the missing numbers that make each set of fractions equal (Equivalent Fractions):

- (a)  $\frac{3}{5} = \frac{\square}{20} = \frac{18}{\square}$ :  
 $20 \div 5 = 4 \implies 3 \times 4 = 12$ .  
 $18 \div 3 = 6 \implies 5 \times 6 = 30$ .  
**Result:**  $\frac{3}{5} = \frac{12}{20} = \frac{18}{30}$ .
- (b)  $\frac{4}{7} = \frac{28}{\square} = \frac{\square}{49}$ :  
 $28 \div 4 = 7 \implies 7 \times 7 = 49$ .  
 $49 \div 7 = 7 \implies 4 \times 7 = 28$ .  
**Result:**  $\frac{4}{7} = \frac{28}{49}$ .
- (c)  $\frac{5}{8} = \frac{15}{\square} = \frac{\square}{64}$ :  
 $15 \div 5 = 3 \implies 8 \times 3 = 24$ .  
 $64 \div 8 = 8 \implies 5 \times 8 = 40$ .  
**Result:**  $\frac{5}{8} = \frac{15}{24} = \frac{40}{64}$ .
- (d)  $\frac{6}{11} = \frac{\square}{33} = \frac{42}{\square}$ :  
 $33 \div 11 = 3 \implies 6 \times 3 = 18$ .  
 $42 \div 6 = 7 \implies 11 \times 7 = 77$ .  
**Result:**  $\frac{6}{11} = \frac{18}{33} = \frac{42}{77}$ .

### Question 4

Write any three equivalent fractions for each of the following:

- (a)  $\frac{2}{3} \cdot \frac{2 \times 2}{3 \times 2} = \frac{4}{6}$ ,  $\frac{2 \times 3}{3 \times 3} = \frac{6}{9}$ ,  $\frac{2 \times 4}{3 \times 4} = \frac{8}{12}$
- (b)  $\frac{4}{5} \cdot \frac{8}{10}, \frac{12}{15}, \frac{16}{20}$
- (c)  $\frac{5}{6} \cdot \frac{10}{12}, \frac{15}{18}, \frac{20}{24}$

### Question 5

From each group given below, identify the set of like fractions:

- (a)  $\frac{2}{7}, \frac{5}{7}, \frac{9}{7}, \frac{3}{7}$ : Like fractions because all denominators are identical (7).
- (b)  $\frac{3}{4}, \frac{3}{5}, \frac{3}{8}, \frac{3}{11}$ : Unlike fractions (same numerator, different denominators).
- (c)  $\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}$ : Like fractions because all denominators are 8.

### Question 6

Check whether the fractions  $\frac{4}{6}$  and  $\frac{20}{30}$  are equivalent or not.

- **Method 1 (Simplification):**  $\frac{4}{6} = \frac{2}{3}$ ,  $\frac{20}{30} = \frac{2}{3}$ .
- **Method 2 (Cross-Multiplication):**  $4 \times 30 = 120$ ;  $6 \times 20 = 120$ .
- **Answer:** Since  $120 = 120$ , the fractions **are equivalent**.

### Question 7

Aarav ate  $\frac{7}{4}$  of a pizza. Write this fraction as a mixed fraction and state whether Aarav ate more than one full pizza or less than one full pizza.

- **Solution:**  $\frac{7}{4} = 1\frac{3}{4}$ .
- Since  $1\frac{3}{4} > 1$ , Aarav ate **more than one full pizza** (one full pizza plus three quarters of another).

### Question 8

Which unit fraction is smaller:  $\frac{1}{9}$  or  $\frac{1}{15}$ ? Give a reason for your answer.

- **Solution:**  $\frac{1}{15}$  is smaller than  $\frac{1}{9}$ .
- \*Reason:\* When 1 whole is divided into 15 equal parts, each part is significantly smaller than when the same whole is divided into only 9 parts. In unit fractions, larger denominator  $\implies$  smaller value.

### Question 9

A tailor has  $\frac{3}{4}$  metre of silk cloth. He wants to write this amount as an equivalent fraction with a denominator of 12 to mark his measurements. What will be the new numerator?

- **Solution:**  $\frac{3}{4} = \frac{x}{12} \implies x = \frac{3 \times 12}{4} = 9$ .
- The new numerator is 9 (fraction is  $\frac{9}{12}$ ).

### Question 10

Sana planned to read  $\frac{3}{5}$  of her book in a day. By evening, she had read  $\frac{12}{20}$  of the book. Did Sana achieve her reading goal? Explain your answer using equivalent fractions.

- **Solution:**

Simplify Sana's progress:  $\frac{12}{20} = \frac{12 \div 4}{20 \div 4} = \frac{3}{5}$ .

Or convert her goal:  $\frac{3 \times 4}{5 \times 4} = \frac{12}{20}$ .

- **Answer: Yes, Sana achieved her reading goal exactly**, because  $\frac{12}{20}$  is mathematically identical and equivalent to  $\frac{3}{5}$ .

## 2. Exercise 8.2: Comparing, Adding & Subtracting Unlike Fractions (Pages 143–144)

### Question 1

Fill in the blanks using the correct sign ( $>$ ,  $<$ , or  $=$ ):

- (a)  $\frac{5}{7} \square \frac{3}{7}$ : Same denominator,  $5 > 3 \implies >$ .
- (b)  $\frac{1}{4} \square \frac{1}{6}$ : Unit fractions,  $4 < 6 \implies >$ .
- (c)  $\frac{3}{8} \square \frac{5}{12}$ : LCM of 8 and 12 is 24.  $\frac{9}{24} < \frac{10}{24} \implies <$ .
- (d)  $\frac{7}{9} \square \frac{14}{18}$ :  $\frac{14}{18} = \frac{7}{9} \implies =$ .

### Question 2

Insert the correct sign ( $<$ ,  $>$ ,  $=$ ):

- (a)  $\frac{2}{3} \square \frac{5}{6} \cdot \frac{4}{6} < \frac{5}{6} \implies <$ .
- (b)  $\frac{7}{10} \square \frac{4}{5} \cdot \frac{7}{10} < \frac{8}{10} \implies <$ .
- (c)  $\frac{5}{12} \square \frac{7}{16}$ : Cross-multiply:  $5 \times 16 = 80$ ;  $12 \times 7 = 84$ . Since  $80 < 84 \implies <$ .

### Question 3

Arrange the following fractions in ascending order:

- (a)  $\frac{2}{3}, \frac{5}{6}, \frac{7}{9}, \frac{1}{2}$ :  
 • Denominators: 3, 6, 9, 2. LCM = 18.  
 • Equivalent fractions:  
 $\frac{2 \times 6}{18} = \frac{12}{18}, \quad \frac{5 \times 3}{18} = \frac{15}{18}, \quad \frac{7 \times 2}{18} = \frac{14}{18}, \quad \frac{1 \times 9}{18} = \frac{9}{18}$ .  
 • Numerator order:  $9 < 12 < 14 < 15$ .  
 • **Ascending Order:**  $\frac{1}{2} < \frac{2}{3} < \frac{7}{9} < \frac{5}{6}$ .
- (b)  $\frac{3}{5}, \frac{7}{10}, \frac{4}{15}, \frac{1}{2}$ :  
 • Denominators: 5, 10, 15, 2. LCM = 30.  
 • Equivalent fractions:  $\frac{18}{30}, \frac{21}{30}, \frac{8}{30}, \frac{15}{30}$ .  
 • **Ascending Order:**  $\frac{4}{15} < \frac{1}{2} < \frac{3}{5} < \frac{7}{10}$ .

### Question 4

Find the sum and write the answer as a mixed fraction:

- (a)  $3\frac{2}{5} + 2\frac{3}{4}$ :  
 • Improper:  $\frac{17}{5} + \frac{11}{4}$ . LCM = 20.  
 $\frac{17 \times 4}{20} + \frac{11 \times 5}{20} = \frac{68+55}{20} = \frac{123}{20} = 6\frac{3}{20}$ .
- (b)  $4\frac{1}{6} + 3\frac{5}{12}$ :  
 • Improper:  $\frac{25}{6} + \frac{41}{12} = \frac{50+41}{12} = \frac{91}{12} = 7\frac{7}{12}$ .
- (c)  $2\frac{3}{8} + 1\frac{5}{6}$ :  
 • Improper:  $\frac{19}{8} + \frac{11}{6}$ . LCM = 24.  
 $\frac{19 \times 3}{24} + \frac{11 \times 4}{24} = \frac{57+44}{24} = \frac{101}{24} = 4\frac{5}{24}$ .

### Question 5

Subtract the following:

- (a)  $\frac{5}{6} - \frac{2}{5}$ : LCM = 30  $\implies \frac{25-12}{30} = \frac{13}{30}$ .
- (b)  $12 - 5\frac{4}{9}$ :  $12 - \frac{49}{9} = \frac{108-49}{9} = \frac{59}{9} = 6\frac{5}{9}$ .
- (c)  $8 - 3\frac{5}{6}$ :  $8 - \frac{23}{6} = \frac{48-23}{6} = \frac{25}{6} = 4\frac{1}{6}$ .

### Question 6

Evaluate:

- (a)  $3\frac{1}{2} - 1\frac{5}{6} + 2\frac{2}{3}$ :  
 $\frac{7}{2} - \frac{11}{6} + \frac{8}{3}$ . LCM = 6.  
 $\frac{21-11+16}{6} = \frac{26}{6} = \frac{13}{3} = 4\frac{1}{3}$ .
- (b)  $2\frac{3}{4} - \frac{1}{2} + 1\frac{1}{8}$ :  
 $\frac{11}{4} - \frac{1}{2} + \frac{9}{8}$ . LCM = 8.

$$\bullet \frac{22-4+9}{8} = \frac{27}{8} = 3\frac{3}{8}.$$

### Question 7

A wire is  $15\frac{3}{5}$  metres long. If a piece of  $6\frac{2}{3}$  metres is cut off, find the length of the wire left.

• **Solution:**

$$\bullet \text{Remaining length} = 15\frac{3}{5} - 6\frac{2}{3} = \frac{78}{5} - \frac{20}{3}.$$

• LCM of 5 and 3 is 15:

$$\frac{78 \times 3 - 20 \times 5}{15} = \frac{234 - 100}{15} = \frac{134}{15} = 8\frac{14}{15} \text{ metres}$$

## 3. Exercise 8.3: Multiplication of Fractions & Lowest Form (Page 146)

### Question 1

**Multiply and simplify to lowest form:**

- (a)  $\frac{2}{3} \times \frac{5}{4}$ :  $\frac{2 \times 5}{3 \times 4} = \frac{10}{12} = \frac{5}{6}$ .
- (b)  $\frac{3}{8} \times \frac{4}{9}$ : Cancel cross factors:  $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ .
- (c)  $2\frac{1}{3} \times 1\frac{1}{5}$ :  $\frac{7}{3} \times \frac{6}{5} = \frac{7 \times 2}{1 \times 5} = \frac{14}{5} = 2\frac{4}{5}$ .
- (d)  $\frac{5}{12} \times \frac{9}{10}$ :  $\frac{1 \times 3}{4 \times 2} = \frac{3}{8}$ .

### Question 2

**Convert into mixed fractions:**

- (a)  $7 \times \frac{2}{3}$ :  $\frac{14}{3} = 4\frac{2}{3}$ .
- (b)  $4 \times \frac{3}{5}$ :  $\frac{12}{5} = 2\frac{2}{5}$ .
- (c)  $5 \times \frac{4}{7}$ :  $\frac{20}{7} = 2\frac{6}{7}$ .

### Question 3

Find the area of a rectangular playground that is  $14\frac{2}{5}$  metres long and  $9\frac{5}{8}$  metres wide.

• **Solution:**

$$\bullet \text{Length} = \frac{72}{5} \text{ m, Width} = \frac{77}{8} \text{ m.}$$

$$\bullet \text{Area} = \frac{72}{5} \times \frac{77}{8} = \frac{9 \times 77}{5 \times 1} = \frac{693}{5} = 138\frac{3}{5} \text{ m}^2 \text{ (or } 138.6 \text{ m}^2).$$

### Question 4

**Find the following:**

- (a)  $\frac{3}{8}$  of 40:  $\frac{3}{8} \times 40 = 3 \times 5 = 15$ .
- (b)  $\frac{2}{3}$  of 1 hour (in minutes):  $\frac{2}{3} \times 60 \text{ min} = 2 \times 20 = 40$  minutes.
- (c)  $\frac{3}{4}$  of 2 kilograms (in grams):  $\frac{3}{4} \times 2000 \text{ g} = 3 \times 500 = 1500$  grams (1.5 kg).

### Question 5

A car runs 16 km using 1 litre of petrol. How much distance will it cover using  $2\frac{3}{4}$  litres of petrol?

• **Solution:**

$$\text{Distance} = 16 \times 2\frac{3}{4} = 16 \times \frac{11}{4} = 4 \times 11 = \mathbf{44 \text{ km}}$$

### Question 6

Rohan has a novel with 120 pages. He reads  $\frac{3}{4}$  of the book.

• (a) What fraction of the book has he not read?

$$1 - \frac{3}{4} = \frac{1}{4}.$$

• (b) How many pages has he read?

$$\frac{3}{4} \times 120 = 3 \times 30 = \mathbf{90 \text{ pages.}}$$

• (c) How many pages remain to be read?

$$120 - 90 = \mathbf{30 \text{ pages.}}$$

### Question 7

A water tanker supplies  $32\frac{1}{2}$  kilolitres of water in a day.  $\frac{2}{5}$  of this water is used for household purposes. Find the amount of water used for household purposes.

• **Solution:**

$$\text{Household water} = \frac{65}{2} \times \frac{2}{5} = \frac{65}{5} = \mathbf{13 \text{ kilolitres}}$$

## 4. Exercise 8.4: Division of Fractions & Reciprocals (Pages 147–148)

### Question 1

Write the reciprocal of each of the following:

• (a)  $\frac{7}{4}$ :  $\frac{4}{7}$

• (b)  $\frac{1}{9}$ :  $9$

• (c)  $2\frac{1}{3}$ : First convert to  $\frac{7}{3} \implies \frac{3}{7}$

• (d)  $6$ :  $\frac{1}{6}$

• (e)  $\frac{5}{8}$ :  $\frac{8}{5}$

### Question 2

Divide the following:

• (a)  $4 \div \frac{1}{2}$ :  $4 \times \frac{2}{1} = \mathbf{8}$ .

• (b)  $\frac{3}{5} \div 6$ :  $\frac{3}{5} \times \frac{1}{6} = \frac{1}{10} = \frac{1}{10}$ .

• (c)  $5\frac{1}{4} \div 2\frac{1}{3}$ :  $\frac{21}{4} \div \frac{7}{3} = \frac{21}{4} \times \frac{3}{7} = \frac{3 \times 3}{4 \times 1} = \frac{9}{4} = \mathbf{2\frac{1}{4}}$ .

• (d)  $\frac{7}{9} \div \frac{14}{27}$ :  $\frac{7}{9} \times \frac{27}{14} = \frac{1 \times 3}{1 \times 2} = \frac{3}{2} = \mathbf{1\frac{1}{2}}$ .

• (e)  $9 \div \frac{3}{4}$ :  $9 \times \frac{4}{3} = 3 \times 4 = \mathbf{12}$ .

### Question 3

A rope of length  $15\frac{3}{4}$  metres is cut into equal pieces of length  $\frac{3}{4}$  metre each. Find the number of pieces.

• Solution:

$$\text{Number of pieces} = 15\frac{3}{4} \div \frac{3}{4} = \frac{63}{4} \times \frac{4}{3} = \frac{63}{3} = 21 \text{ pieces}$$

### Question 4

A shopkeeper has 27 kg of dry fruits. He distributes them equally among 36 children. How many grams of dry fruits does each child receive?

• Solution:

- Total mass in grams =  $27 \times 1000 = 27,000$  g.
- Each child receives =  $\frac{27000}{36} = 750$  grams (or  $\frac{3}{4}$  kg).

### Question 5

A garden plot is divided into 4 equal sections.  $\frac{2}{3}$  of each section is planted with flowers. What fraction of the entire garden is planted with flowers? If the area of the garden is  $240 \text{ m}^2$ , what is the planted area?

• Solution:

- Fraction of whole garden =  $4 \times (\frac{1}{4} \times \frac{2}{3}) = \frac{2}{3}$ .
- Planted area =  $\frac{2}{3} \times 240 \text{ m}^2 = 160 \text{ m}^2$ .

### Question 6

The cost of  $7\frac{1}{2}$  kg of sugar is ₹340. Find the price of 1 kg of sugar.

• Solution:

$$\text{Price per kg} = 340 \div 7\frac{1}{2} = 340 \div \frac{15}{2} = 340 \times \frac{2}{15} = \frac{680}{15} = ₹45.33$$

(or ₹45 $\frac{1}{3}$ ).

### Question 7

A wire of length  $18\frac{3}{4}$  metres is distributed equally among 15 students for a science project. Find the length of wire received by each student.

• Solution:

$$\text{Length per student} = \frac{75}{4} \div 15 = \frac{75}{4} \times \frac{1}{15} = \frac{5}{4} = 1\frac{1}{4} \text{ metres}$$

### Question 8

A tailor uses  $2\frac{1}{4}$  metres of cloth to stitch one shirt. How many such shirts can be stitched from a roll of  $33\frac{3}{4}$  metres?

• Solution:

$$\text{Number of shirts} = 33\frac{3}{4} \div 2\frac{1}{4} = \frac{135}{4} \div \frac{9}{4} = \frac{135}{4} \times \frac{4}{9} = \frac{135}{9} = 15 \text{ shirts}$$

### Question 9

Mira is reading a novel that has 480 pages. She read  $\frac{5}{8}$  of the pages today. How many more pages does she need to read to complete the book?

• **Solution:**

- Fraction remaining  $= 1 - \frac{5}{8} = \frac{3}{8}$ .
- Pages remaining  $= \frac{3}{8} \times 480 = 3 \times 60 = 180$  pages.

### Question 10

A car travels a distance of  $202\frac{1}{2}$  km in  $4\frac{1}{2}$  hours. What is the average speed of the car in km/h?

• **Solution:**

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{405}{2} \div \frac{9}{2} = \frac{405}{2} \times \frac{2}{9} = \frac{405}{9} = 45 \text{ km/h}$$

## 5. Chapter 8 Drill: Complete Question-by-Question Solutions (Pages 150–153)

### Section A: Multiple Choice Questions (MCQs 1–10)

1. Which of the following is a unit fraction?

- \*(a)  $\frac{4}{1}$  | (b)  $\frac{1}{7}$  | (c)  $\frac{7}{4}$  | (d)  $\frac{4}{7}$ \*

• **Correct Option: (b)  $\frac{1}{7}$ .**

2. The reciprocal of  $2\frac{1}{3}$  is:

- \*(a)  $\frac{7}{3}$  | (b)  $\frac{3}{7}$  | (c)  $\frac{1}{7}$  | (d)  $2\frac{3}{1}$ \*

• **Correct Option: (b)  $\frac{3}{7}$ .**

3. To compare  $\frac{2}{5}$  and  $\frac{3}{4}$ , the common denominator should be:

- \*(a) 5 | (b) 4 | (c) 20 | (d) 10\*

• **Correct Option: (c) 20.**

4. A fraction is in its simplest form if its numerator and denominator are:

- \*(a) Equal | (b) Co-prime | (c) Prime numbers | (d) Even numbers\*

• **Correct Option: (b) Co-prime.**

5. What is  $\frac{3}{4}$  of 16?

- \*(a) 12 | (b) 8 | (c) 4 | (d) 16\*

• **Correct Option: (a) 12.**

6. Which symbol makes the statement true:  $\frac{3}{4} \square \frac{9}{12}$ ?

- \*(a)  $>$  | (b)  $<$  | (c)  $=$  | (d)  $\leq$ \*

• **Correct Option: (c)  $=$ .**

7. An improper fraction is always:

- \*(a) Less than 1 | (b) Greater than or equal to 1 | (c) Equal to 1 | (d) A unit fraction\*

• **Correct Option: (b) Greater than or equal to 1.**

8.  $4 \div \frac{1}{2}$  is equal to:

\*(a) 2 | (b) 8 | (c)  $\frac{1}{8}$  | (d) 4\*

• **Correct Option: (b) 8.**

9. An equivalent fraction of  $\frac{2}{3}$  with denominator 12 is:

\*(a)  $\frac{4}{12}$  | (b)  $\frac{8}{12}$  | (c)  $\frac{6}{12}$  | (d)  $\frac{10}{12}$ \*

• **Correct Option: (b)  $\frac{8}{12}$ .**

10. If  $\frac{5}{8} = \frac{15}{x}$ , then the value of  $x$  is:

\*(a) 16 | (b) 24 | (c) 30 | (d) 40\*

• **Correct Option: (b) 24.**

## Section B: Fill in the Blanks (Q1–Q6)

1. The number that does not have a reciprocal is 0.
2. In the fraction  $\frac{7}{11}$ , the top number 7 is called the **numerator**.
3. Fractions with the same denominator are called **like** fractions.
4. While dividing fractions, we multiply by the **reciprocal** of the divisor.
5. The product of  $\frac{3}{5} \times \frac{5}{3}$  is 1.
6. A fraction written as  $3\frac{1}{2}$  is called a **mixed** fraction.

## Section C: Very Short Answer Questions (Q1–Q11)

1. **Define a proper fraction and give one example.**

• A fraction whose numerator is strictly less than its denominator (e.g.,  $\frac{3}{5}$ ).

2. **Convert  $\frac{22}{4}$  into a mixed fraction.**

•  $\frac{22}{4} = \frac{11}{2} = 5\frac{1}{2}$ .

3. **Find the HCF of 24 and 36, and use it to simplify  $\frac{24}{36}$ .**

•  $\text{HCF}(24, 36) = 12$ .

•  $\frac{24 \div 12}{36 \div 12} = \frac{2}{3}$ .

4. **Write the reciprocal of  $3\frac{1}{5}$ .**

•  $3\frac{1}{5} = \frac{16}{5} \implies \text{Reciprocal is } \frac{5}{16}$ .

5. **Is  $\frac{12}{12}$  a proper fraction or an improper fraction? Give a reason.**

• **Improper fraction.** In a proper fraction, the numerator must be strictly less than the denominator. When numerator  $\geq$  denominator, it is improper.

6. **Multiply  $9 \times \frac{4}{27}$ .**

•  $9 \times \frac{4}{27} = \frac{1 \times 4}{3} = \frac{4}{3} = 1\frac{1}{3}$ .

7. **Write the next two equivalent fractions for  $\frac{3}{7}$ .**

•  $\frac{6}{14}$  and  $\frac{9}{21}$ .

8. **Compare  $\frac{5}{8}$  and  $\frac{7}{12}$  using any suitable method.**

• Cross-multiplication:  $5 \times 12 = 60$ ;  $8 \times 7 = 56$ . Since  $60 > 56 \implies \frac{5}{8} > \frac{7}{12}$ .

9. How many  $\frac{2}{3}$  kg pieces can be cut from a cake of weight 4 kg?

•  $4 \div \frac{2}{3} = 4 \times \frac{3}{2} = 6$  pieces.

10. Divide  $\frac{5}{8}$  by  $(\frac{3}{4} \text{ of } \frac{5}{6})$ .

•  $\frac{3}{4} \times \frac{5}{6} = \frac{5}{8}$ .

•  $\frac{5}{8} \div \frac{5}{8} = 1$ .

11. Which is heavier:  $\frac{12}{25}$  of 500 grams or  $\frac{3}{20}$  of 4 kg?

• Quantity 1:  $\frac{12}{25} \times 500 \text{ g} = 12 \times 20 = 240$  grams.

• Quantity 2:  $\frac{3}{20} \times 4000 \text{ g} = 3 \times 200 = 600$  grams.

• Answer:  $\frac{3}{20}$  of 4 kg (600 g) is heavier.

### Section D: Short Answer Questions (Q1–Q6)

1. Simplify  $\frac{72}{108}$  using the cancellation method, showing all steps.

•  $\frac{72 \div 36}{108 \div 36} = \frac{2}{3}$ .

2. Arrange the fractions  $\frac{2}{3}, \frac{5}{6}, \frac{7}{12}$  in ascending order.

• LCM = 12  $\implies \frac{8}{12}, \frac{10}{12}, \frac{7}{12}$ .

• Order:  $\frac{7}{12} < \frac{2}{3} < \frac{5}{6}$ .

3. A rope is  $\frac{3}{4}$  metre long. How many such ropes are required to make 6 metres?

•  $6 \div \frac{3}{4} = 6 \times \frac{4}{3} = 8$  ropes.

4. Find the value of  $x$  if  $\frac{x}{7} = \frac{15}{35}$ .

•  $\frac{15}{35} = \frac{3}{7} \implies x = 3$ .

5. Convert the mixed fraction  $7\frac{3}{8}$  into an improper fraction and then find its reciprocal.

• Improper:  $\frac{59}{8}$ . Reciprocal:  $\frac{8}{59}$ .

6. Solve:  $(2\frac{1}{2} \times 1\frac{1}{5}) \div \frac{3}{4}$ .

•  $(\frac{5}{2} \times \frac{6}{5}) \div \frac{3}{4} = 3 \div \frac{3}{4} = 3 \times \frac{4}{3} = 4$ .

### Section E: Long Answer Questions (Q1–Q10)

1. The area of a rectangular plot is  $65\frac{1}{3} \text{ m}^2$ . If its breadth is  $5\frac{1}{3}$  metres, find its length.

• Length = Area  $\div$  Breadth =  $\frac{196}{3} \div \frac{16}{3} = \frac{196}{3} \times \frac{3}{16} = \frac{196}{16} = 12\frac{1}{4}$  metres (12.25 m).

2.  $18\frac{3}{4}$  litres of juice is poured into small bottles of capacity  $\frac{3}{4}$  litre each. How many bottles can be filled completely?

• Bottles =  $\frac{75}{4} \div \frac{3}{4} = \frac{75}{3} = 25$  bottles.

3. Mili reads  $\frac{1}{3}$  part of a book in 1 hour. How much part of the book will she read in  $2\frac{1}{5}$  hours?

• Part read =  $\frac{1}{3} \times 2\frac{1}{5} = \frac{1}{3} \times \frac{11}{5} = \frac{11}{15}$  part.

4. Vidya and Pratap went for a picnic. Their mother gave them a water bottle that contained 5 litres of water. Vidya consumed  $\frac{2}{5}$  of the water. Pratap consumed the remaining water.

• \*(a) How much water did Vidya drink?  $\frac{2}{5} \times 5 = 2$  litres.

• \*(b) What fraction of the total quantity did Pratap drink?  $1 - \frac{2}{5} = \frac{3}{5}$  (which is 3 litres).

5. Saili plants 4 saplings in a row in her garden. The distance between two adjacent saplings is  $\frac{3}{4}$  metre. Find the distance between the first and the last sapling.

• Number of intervals between 4 saplings =  $4 - 1 = 3$ .

• Total distance =  $3 \times \frac{3}{4} = \frac{9}{4} = 2\frac{1}{4}$  metres (2.25 m).

**6. In a class of 40 students,  $\frac{1}{5}$  of the total students like to study English,  $\frac{2}{5}$  of the total like to study Mathematics, and the remaining students like to study Science.**

• \*(a) How many students like to study English? \*  $\frac{1}{5} \times 40 = 8$  students.

• \*(b) How many students like to study Mathematics? \*  $\frac{2}{5} \times 40 = 16$  students.

• \*(c) What fraction of the total students like to study Science? \*  $1 - \left(\frac{1}{5} + \frac{2}{5}\right) = 1 - \frac{3}{5} = \frac{2}{5}$  (which is 16 students).

**7. A cement company earns a profit of ₹8 per bag of white cement sold and a loss of ₹5 per bag of grey cement sold. If it sells 3,000 bags of white cement and 5,000 bags of grey cement in a month, what is its profit or loss?**

• Profit =  $3000 \times 8 = ₹24,000$ .

• Loss =  $5000 \times 5 = ₹25,000$ .

• Net result =  $24,000 - 25,000 = -₹1,000 \implies$  **Net Loss of ₹1,000.**

**8. Evaluate:**  $\left(\frac{2}{3} + \frac{1}{2}\right) \times \left(\frac{5}{6} - \frac{1}{4}\right) \div \frac{7}{12}$ .

• Term 1:  $\frac{4+3}{6} = \frac{7}{6}$ .

• Term 2:  $\frac{10-3}{12} = \frac{7}{12}$ .

• Product:  $\frac{7}{6} \times \frac{7}{12} = \frac{49}{72}$ .

• Division:  $\frac{49}{72} \div \frac{7}{12} = \frac{49}{72} \times \frac{12}{7} = \frac{7}{6} = 1\frac{1}{6}$ .

**9. A rectangular field is  $24\frac{1}{2}$  m long and  $15\frac{2}{3}$  m broad. Find its perimeter and cost of fencing at ₹20 per metre.**

• Perimeter =  $2 \times (l + b) = 2 \times \left(\frac{49}{2} + \frac{47}{3}\right) = 2 \times \frac{147+94}{6} = 2 \times \frac{241}{6} = \frac{241}{3} = 80\frac{1}{3}$  m.

• Cost of fencing =  $\frac{241}{3} \times 20 = \frac{4820}{3} = ₹1606.67$ .

**10. A family consumes  $3\frac{1}{2}$  kg of rice in a week. How much rice will they consume in  $4\frac{2}{7}$  weeks?**

• Total consumption =  $3\frac{1}{2} \times 4\frac{2}{7} = \frac{7}{2} \times \frac{30}{7} = \frac{30}{2} = 15$  kg.

## Section F: HOTS & Life Skills (Page 153)

**1. Shade  $\frac{1}{2}$  of a rectangle. Now, divide the shaded part into 3 equal parts and take one part. What fraction of the original rectangle is this?**

• Calculation:  $\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$ .

**2. What is the reciprocal of  $\frac{1}{5}$ ?**

• 5.

**3. Which is greater:  $\frac{1}{2}$  of 20 or  $\frac{1}{4}$  of 40?**

•  $\frac{1}{2} \times 20 = 10$ ;  $\frac{1}{4} \times 40 = 10$ .

• **Answer: Both are equal ( $10 = 10$ ).**

**4. If you have  $\frac{3}{4}$  of a pizza and share it equally among 3 people, what fraction of a whole pizza does each person receive?**

•  $\frac{3}{4} \div 3 = \frac{3}{4} \times \frac{1}{3} = \frac{1}{4}$  of the pizza.

**5. How does sharing resources (like fractional portions of food, time, or knowledge) build empathy and community cooperation?**

- When we share fractional portions, we recognize that no individual requires the entire whole to thrive. Distributing parts equitably ensures everyone's needs are met, mirroring the harmony of common denominators working in unison.